WEEKLY ROLLUP • PhD RESEARCH WATCH

Particulate Matter
Week in Review

Week ending Saturday 1 August 2026 | pooled 27 July 2026 to 1 August 2026 | 131 records from 6 daily issues

131 RECORDS POOLED THIS WEEK	**20** SENSING RECORDS (15% OF WEEK)	**10** SUBTOPICS ACTIVE	**48** POOLED EFFECT ESTIMATES	**4** TRIALS ON THE WATCH LIST

The week in one paragraph

Four independent papers this week attacked the exposure *variable* rather than the health model — and all four found it wanting. The through-line is that $PM_{2.5}$ mass at a fixed monitor is now the weakest link in the causal chain, and the field has started saying so with numbers.

Nyoni et al. (31 Jul, *Air Qual Atmos Health*) validated road-geometry proxies against a year of filter-based personal $PM_{2.5}$ in 343 postpartum participants across three sub-Saharan countries: weighted road density correlated only in Mozambique ($\rho = 0.351$) and **distance-to-highway inverted sign in Kenya** ($\rho = -0.224$, $p = 0.018$). Saeedi et al. (1 Aug, *ES&T*) showed land-use regression for mobile ultrafine particles falling from **~217% average percentage error under random cross-validation to ~79% under spatiotemporal CV** — the model was never predicting, the validation was. Blanco et al. (1 Aug, *JESEE*) resampled on-road campaign designs against a 5,283-person cohort and found every alternative design attenuated the health estimate. And Zhu et al. (31 Jul) reported pneumonia-admission RRs of **1.004 (0.991–1.017) for $PM_{2.5}$ against 1.221 (1.174–1.271) for NO_2** — PM mass was the null term in its own model.
What would have to be true for this to be wrong: the four results would have to be selection rather than signal. Two of them (Saeedi, Blanco) are methods papers whose premise is that current practice is inadequate, so they are not independent of the conclusion; Nyoni's setting is one where household biomass dominates the personal exposure budget, which is exactly where traffic geometry should fail; and Zhu's is a single-pollutant model over correlated combustion tracers, so it is a statement about correlation structure, not about which pollutant acts. **This is a real pattern, but it is a pattern about measurement adequacy, not about PM being harmless.** The distinction matters and the week's own papers do not always draw it.

Subtopic trend and inflection

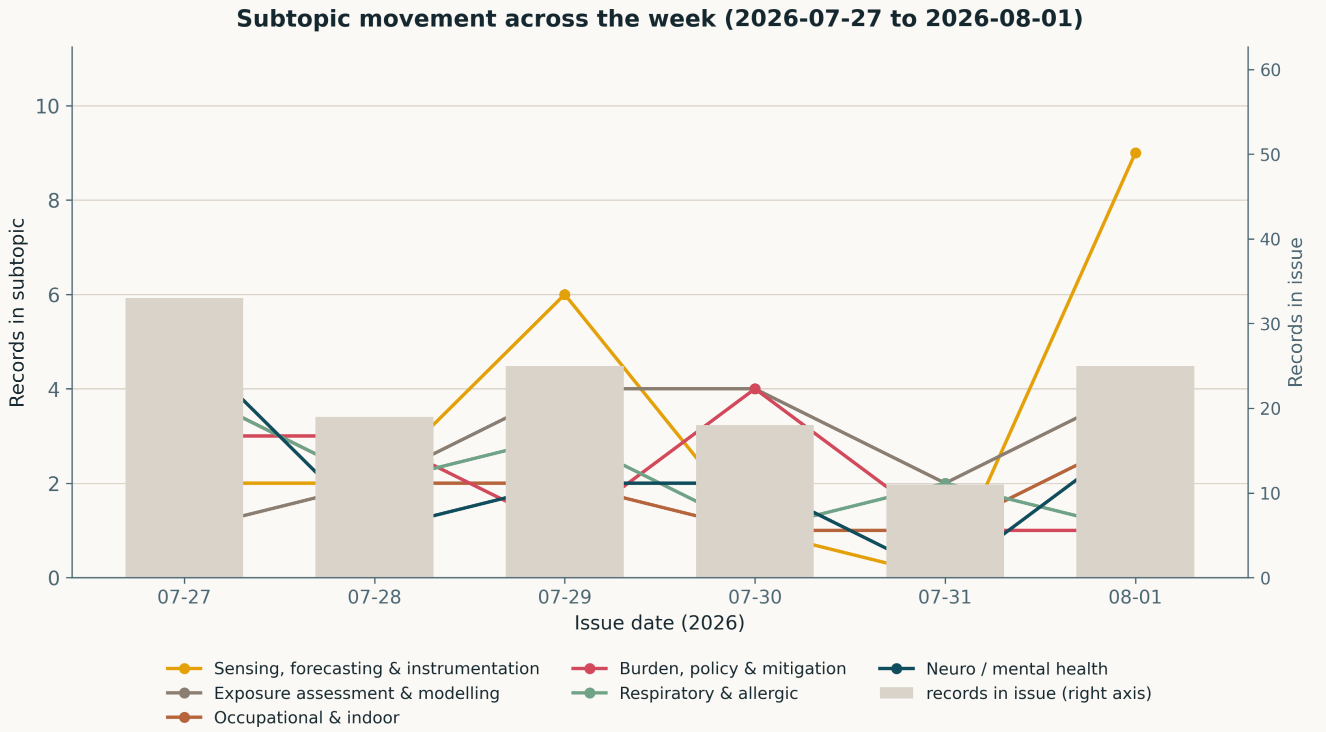

Fig. Per-issue subtopic counts across the week from `state/metrics.csv`. **The one real inflection is sensing/instrumentation: 2, 2, 6, 1, 0, 9 across 27 July–1 August, against daily totals of 33, 19, 25, 18, 11, 25.** Both ends of that series are pipeline artefacts, not literature. The 6 on 29 July came from a Consensus backfill; the 0 on 31 July and the 9 on 1 August are the same defect and its fix — the Crossref by-journal leg was activated on 1 August and back-swept the week, so nine records that were deposited between 26 July and 1 August all landed in one issue. **Read the weekly total (20/131, 15%) rather than any daily value.** Every other subtopic is flat within Poisson noise on a base of 11–33 records/day; nothing else in this figure is a trend.

Emerging this week. Three genuinely new lines. (i) **Source-resolved exposure** — Dales et al. separate wildfire-sourced from background $PM_{2.5}$ and find +6.07% (5.75–6.39) headache ED visits for the former against a null for the latter; Zhou M et al. attribute 27% of India's population-weighted $PM_{2.5}$ to transboundary transport, more than any domestic sector. (ii) **Ultrafine and number-based metrics** — 6 of 131 records this week use UFP or particle-number exposure (Ramsden, Zhou 2026b, Rusconi, Zhou L, Saeedi, Blanco), against essentially none in the preceding month's dailies. (iii) **Oxidative potential as a modelled quantity** — Mishra et al.'s KM-OP predicts OP from composition at $R^2 > 0.75$, which is the step that makes speciated sensing actionable.

Fading. Nothing has genuinely faded in six days, and claiming otherwise would be noise-fitting. The apparent drop in cardiovascular/metabolic records (4 on 29 July, 2 on 30 July, 0 on 31 July and 1 August) is journal-issue timing on an 11–25-record base, not a decline.

Flat but load-bearing. Exposure assessment & modelling (17 records) and mechanistic toxicology (11) held steady all week and carry the field regardless of the weekly headline. Reproductive/developmental (10) is the cluster with the most consistent effect estimates — Xie's pubertal-timing HRs, Li's stillbirth HRs, Sieber's LBW OR and Pai's incense aORs are all in the same direction with intervals that exclude the null.

Pooled analytics

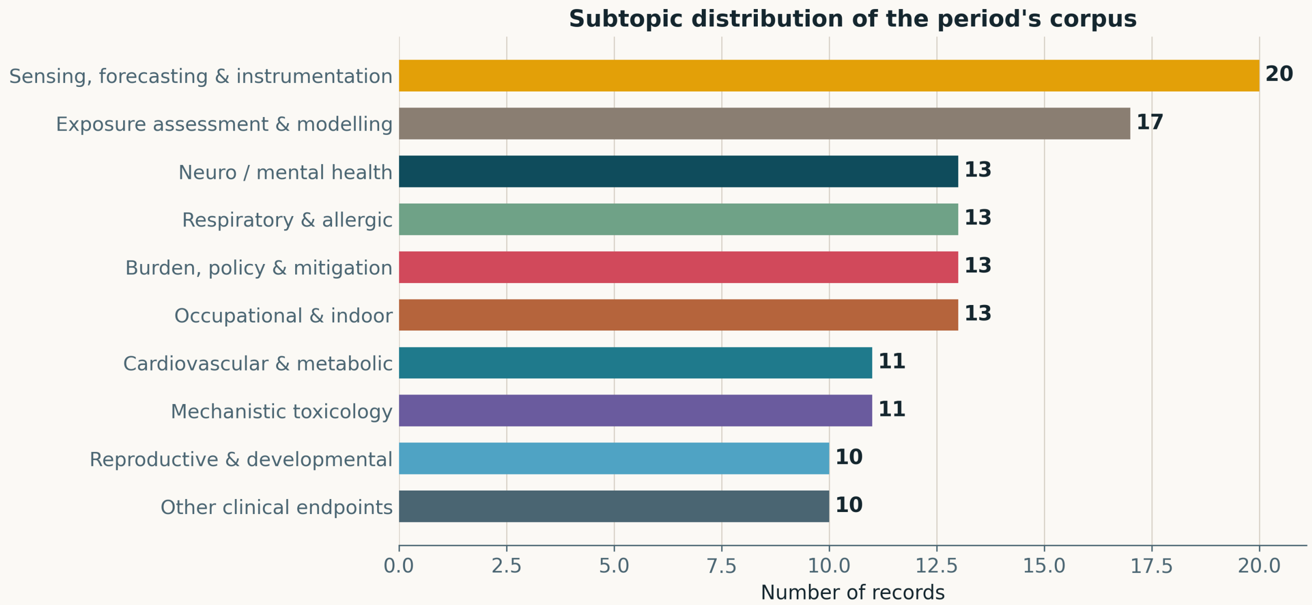

Fig. Pooled subtopic distribution, 27 July 2026 to 1 August 2026. Sensing (20) and exposure assessment (17) together account for **28% of the week** — the methodological half of the field is, for once, better represented than any single health cluster. Neuro/mental health, respiratory/allergic, burden/policy and occupational/indoor tie at 13.

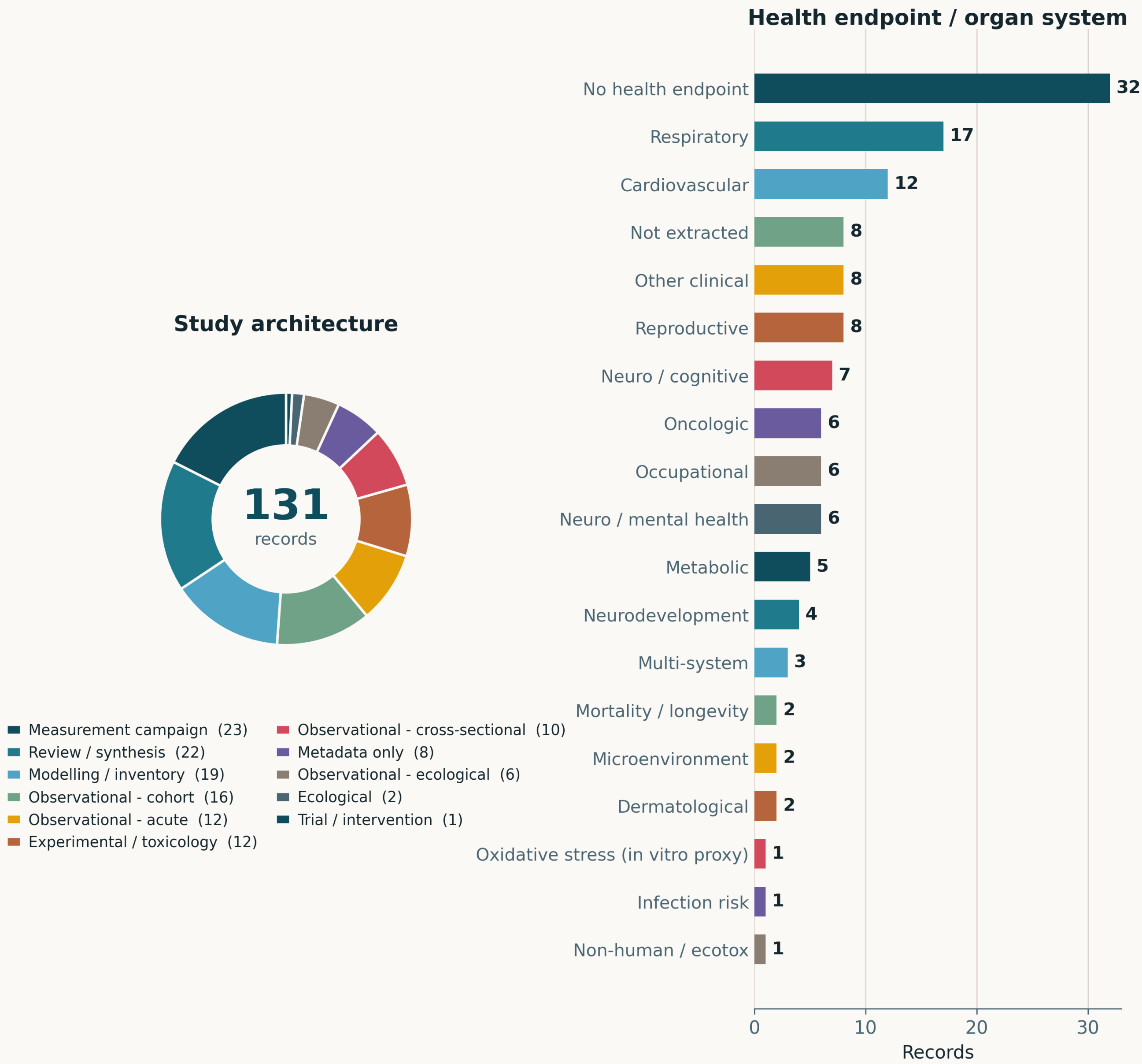

Fig. Pooled study architecture and health endpoint. Measurement campaigns lead at 23, but **reviews and syntheses are effectively tied at 22 — one record in six of this week's corpus is a review**, a high share that reflects a mid-year cluster of review issues rather than new primary evidence. Observational designs together contribute 36 (16 cohort, 12 acute, 6 cross-sectional, 2 ecological) and only one record is a trial or intervention. **Thirty-two records carry no health endpoint** and eight more were surfaced without an abstract; among those that do, respiratory (17) and cardiovascular (12) lead.

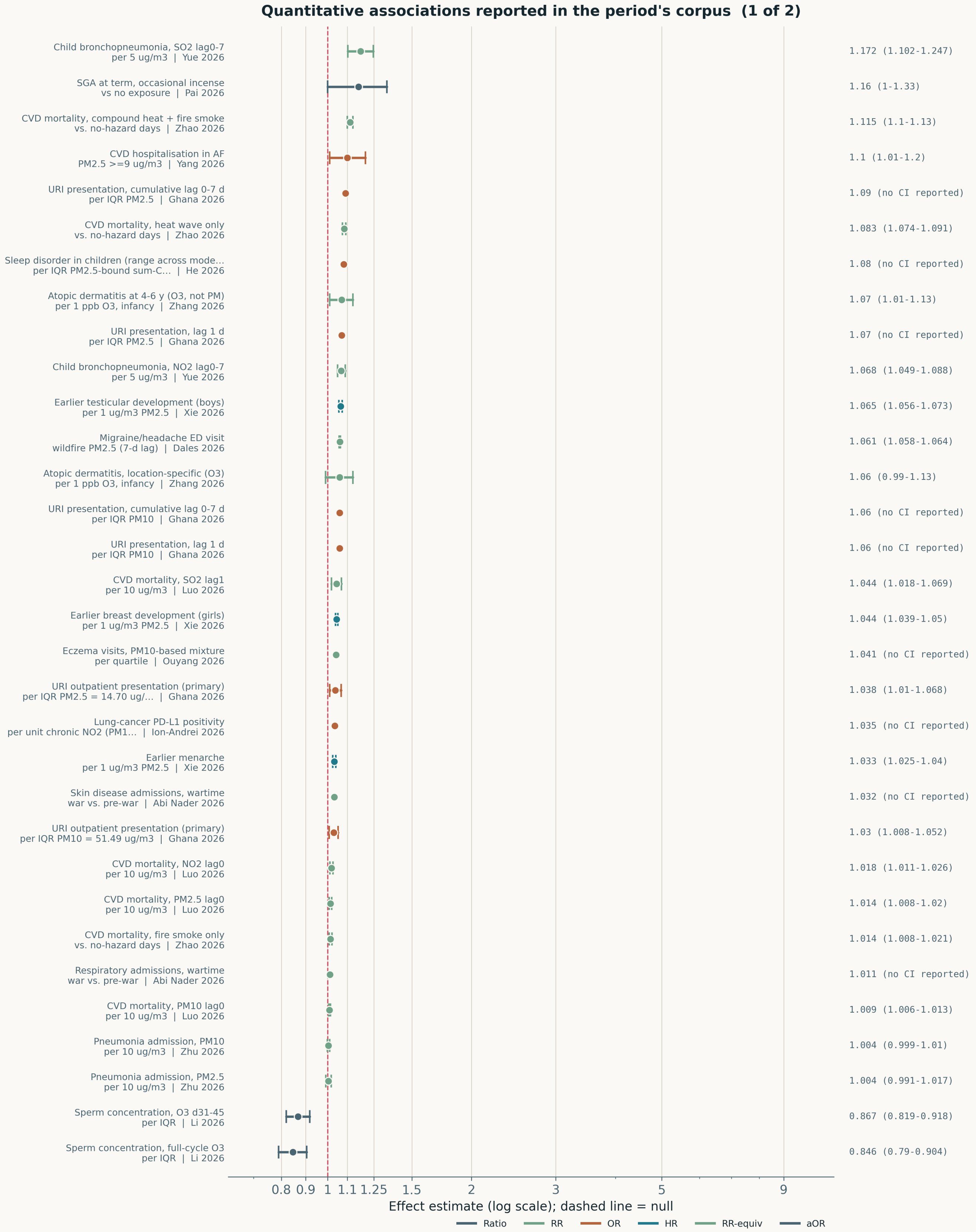

Fig. Pooled effect estimates for the week — read as a caterpillar plot, not a meta-analysis. No summary diamond is drawn and none should be. **Pooling is not defensible here and no I^2 is reported**, for four separate reasons: exposure contrasts differ (per 1 $\mu g\,m^{-3}$ in Xie, per IQR in Sánchez, a $\geq 9\,\mu g\,m^{-3}$ threshold in Yang, a percentage-change conversion in Dales); outcomes differ across ten organ systems; several estimates are multiple contrasts from one paper and one cohort (Yang contributes 3, Xie 3, Ghana AIRCLARE 6, Luo 4), so they are not independent; and 7 of the 48 estimates have **no published interval at all** (Mokoena $\times 3$, Ouyang $\times 2$, Abi Nader $\times 2$) and are plotted as points. The one subset that is internally comparable is the four respiratory RRs per 10 $\mu g\,m^{-3}$ from Zhu and Yue — and within that subset the particulate terms (1.004, 1.004) are null while the gas terms (1.221, 1.068, 1.172) are not.

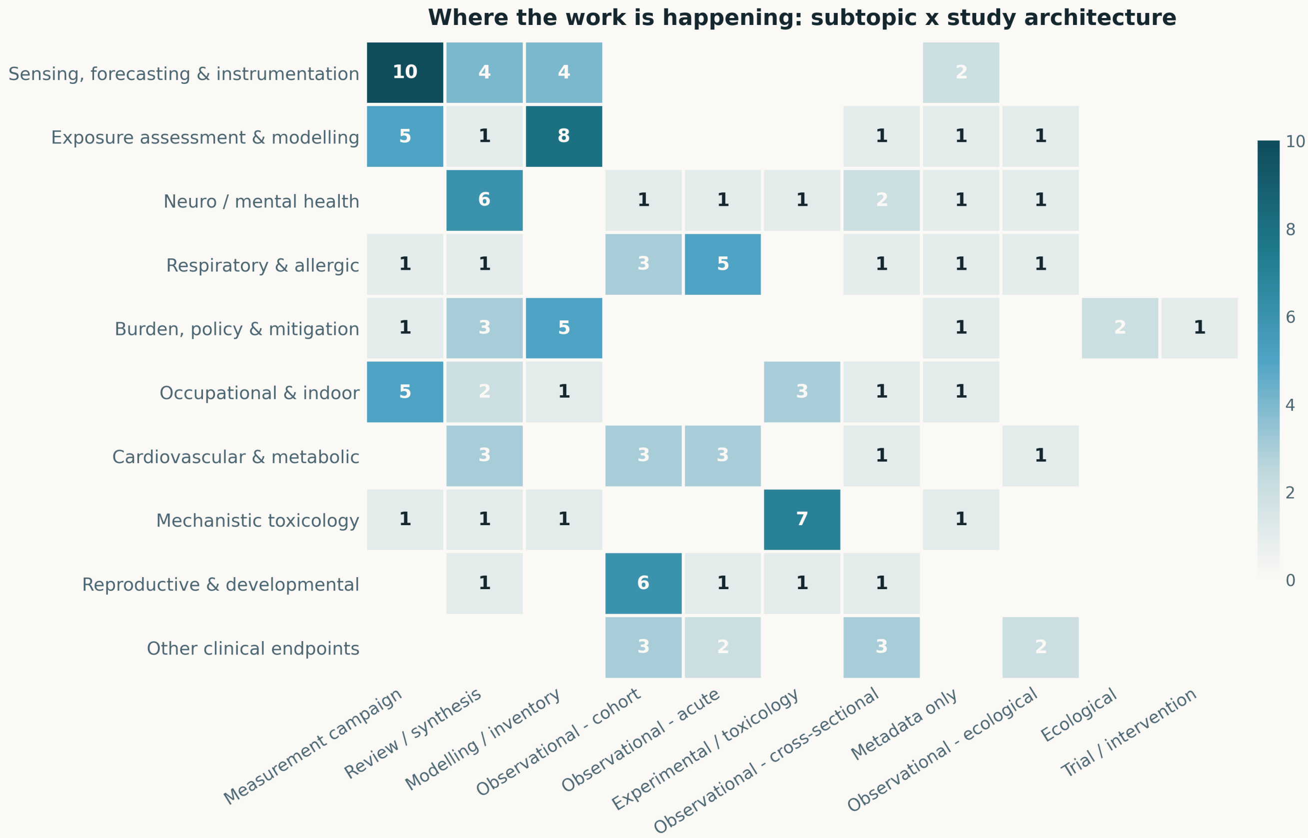

Fig. Pooled subtopic × study-architecture cross-tabulation. **The persistently empty cells are the finding.** Mechanistic toxicology has no cell in any observational column — eleven records, no bridge to human biomarker work. The sensing row has no cell in any cohort or case-crossover column: across 131 records and six days, **no study characterised an instrument and used it to estimate a health effect in the same paper.** Eight records come closest (Wang's SARS-CoV-2 surveillance, Patton's outdoor risk assessment, Sánchez's asthma admissions, Chen's 29-hazard mixture, Blanco's UFP–cognition analysis) but in every case the exposure instrument is inherited, not evaluated.

Groups, venues and most-cited work

Recurring author and group clusters. There are none, and the apparent ones are an artefact worth naming. Surname frequency over the pooled corpus returns Chen (5), Zhou (5), Ding, Park, Lu, Wang, Zhang and Li (3 each) — but these are distinct research groups sharing common surnames, not repeat output. The 1 August issue alone carries two unrelated first-author Zhous (Liyuan Zhou on Bakhoor emissions, Mi Zhou on India source attribution), which is why they are disambiguated as "Zhou L" and "Zhou M" in the register. **First-author surname is not a group identifier and should not be used as one**; a real cluster analysis needs ORCID or affiliation strings, which this pipeline does not currently store. That is a state-schema gap, logged as such. The only genuine programmatic stream visible is the Henan Rural Cohort (Zhang, 1 Aug) and the Chongqing Child Health Cohort (Xie, 27 Jul), each contributing one record.

Venue concentration. Nineteen journals contributed more than one record. *Toxics* leads at 7, *J Environ Sci* (*China*), *J Hazard Mater* and *Atmos Environ* at 6 each, *Front Public Health* and *Environ Sci Pollut Res* at 3, *Atmos Chem Phys* at 3. Two of these are structural rather than substantive: the *Atmos Environ* count is entirely the Crossref by-journal leg activated on 1 August and reflects a deposit window, not a research surge; the *Toxics* cluster is a themed-issue effect. **No single-journal cluster this week represents an independent replication.**

Most-cited work of the period. Not reported, because it would be noise. Every record in this corpus entered an index within the last six days; citation counts where available are 0–5 and are dominated by how long a preprint sat before formal publication, not by influence. Ranking a week-old corpus by citations measures publication lag. The metric becomes meaningful in the monthly and yearly rollups and is deferred to them.

Trial registry watch

From `state/trials.json`: registrations created or updated inside the window. Daily issues accumulate; the rollup is where they surface.

New this week — three sham-controlled filtration trials (surfaced 1 August on an intervention sweep; all three are indoor HEPA filtration against a sham unit, which is the correct comparator and rules out the placebo-and-noise objection that sinks open-label purifier studies).

- **NCT05867381** — *Slowing Atherothrombosis Progression Through Indoor Air Filtration*, University of Southern California. Recruiting, N/A phase, **n = 112**, crossover, HEPA vs sham, adults with ischaemic heart disease history, primary completion **June 2028**. Crossover on a progression endpoint is the right design for a within-person exposure contrast, but 112 participants against an atherothrombosis imaging endpoint is thin: the indoor $PM_{2.5}$ reduction a HEPA unit delivers (∼30–60% of indoor mass) implies an effect far smaller than the between-person variance in plaque progression. Expect a surrogate-endpoint result.
- **NCT05718245** — *Using Indoor Air Filtration to Reduce $PM_{2.5}$ Cardiometabolic Effects in At-risk Individuals*, USC. Recruiting, **n = 52**, HEPA vs sham, primary completion **December 2026** — the nearest-term readout on the list. At $n = 52$ this is powered for biomarker change (blood pressure, glucose, inflammatory markers), not for events, and should be read that way.
- **NCT05016271** — *Health Benefits of Air Purifiers in Primary School Students*, Fudan University. Active, not recruiting, **n = 110**, HEPA-equipped vs non-HEPA ventilation in classrooms, cardiopulmonary function. **Primary completion was June 2022 and no results have been posted** — four years is well past the reporting expectation and is itself worth logging.

Is the endpoint capable of detecting the effect? For all three: not for hard outcomes. A sham-controlled filtration trial changes personal $PM_{2.5}$ by single-digit $\mu g\, m^{-3}$ over weeks to months; the cohort literature in this same week (Luo: CVD-mortality RR 1.014 per 10 $\mu g\, m^{-3}$; Yang: OHCA RR 1.02) implies effect sizes that a 52–112-person trial cannot resolve on events. **These trials are mechanism studies wearing RCT clothing**, and their value is the biomarker and personal- exposure data, not the clinical endpoint.

Standing watch list. NCT05874479 — *Reducing Air Pollution Exposure to Lower Blood Pressure Among NYC Public Housing Residents* (NYU Langone, $n = 440$, active vs sham portable air cleaner, primary completion September 2027). Logged 29 July; at $n = 440$ it is the only trial on the list plausibly powered for a blood-pressure endpoint. Nearest readout to watch: **NCT05718245, December 2026**.

Open gaps

- **Measurement–epidemiology coupling: still zero.** Across 131 records, no study both characterised an instrument and estimated a health effect with it. Blanco et al. came closest — a mobile UFP campaign feeding a cognition analysis in the same paper — but the monitoring instrument itself is not evaluated, only the campaign design. This is the same gap the 27 July issue named and it has not moved.
- **Ultrafine and speciated metrics: improving, from nothing.** 6 of 131 records (4.6%) use UFP or particle-number exposure and 3 use speciated composition. That is up from near-zero, entirely because the aerosol-science journals are now being read. It remains badly out of proportion to the deposition physics: the fraction of deposited particle *number* that reaches the alveolar region is dominated by exactly the size range that 95% of this week's records do not measure.
- **Geographic coverage vs. attributable burden: the gap narrowed.** 13 of 131 records (10%) come from South Asia or sub-Saharan Africa — India 7, Ghana, Uganda, South Africa, Bangladesh and a three-country Gambia/Kenya/Mozambique study. Against China's 42 (32%) this is still inverted relative to attributable burden, but it is the best week on this measure so far, and two of the records (Nyoni, Ghana AIRCLARE) are primary measurement rather than secondary analysis.
- **Calibration transferability: reported, and it fails.** Three records tested transfer explicitly. Ginsburg et al. (29 Jul) found US-derived PurpleAir corrections do not transfer outside North America across 1,013 globally co-located sensors, with region-specific corrections needed to meet EPA 24 h targets. Kumpika et al. report $R^2 = 0.974$ but on single-site, single-season, single-model data — co-located fit, not held-out deployment. Qian et al. report chamber performance only. **No record this week reported held-out deployment performance at a site not used in fitting.**
- **Negative and null results: present, which is a good sign.** Zhu et al.'s null $PM_{2.5}$ pneumonia RR, Blanco et al.'s null UFP–cognition estimate, Yang et al.'s null PM$\times$heat interaction and the AFRI-c filtration process evaluation all published as nulls. A corpus with no nulls would be a reporting-bias signal; this one has four in six days, which is consistent with a literature that is at least sometimes publishing what it finds.

Where this is heading — judgement, not measurement

The methodological demand the field is converging on is **source- and size-resolved exposure with a validated error term**. Three separate literatures arrived at it independently this week: epidemiology (Dales's wildfire-vs-background split, Zhu's null mass term), exposure modelling (Saeedi and Blanco on validation and design), and aerosol science (Zhou L on number-dominated indoor emissions, Mishra on predicting oxidative potential from composition). None of them cites the others. That convergence is the strongest signal in the week and I would put moderate-to-high confidence on it persisting.

What would change the picture: a well-powered cohort finding a robust total-mass association after adjusting for source and composition would undercut the whole line, and none of this week's records tested that. The mass-null results here come from single-pollutant models over correlated combustion tracers, which is a weak basis for demoting mass. **I would not yet write "$PM_{2.5}$ mass is the wrong metric" in a thesis introduction** — I would write that its measured association is increasingly sensitive to source attribution and model specification, which is a defensible and much narrower claim.

For a distributed low-cost PM sensing programme the operational reading is specific. First, **number-based channels are now worth the extra cost**: the week produced two independent results (Bakhoor UFP emission rates exceeding sidestream cigarette smoke; 6 UFP records against essentially zero the previous month) that a mass-only network cannot see. Second, **validate on held-out sites, not held-out samples** — Saeedi's 217%→79% collapse is what random-CV validation costs, and it applies directly to any network calibration reported as "R^2 against co-located reference". Third, **instrument ageing and firmware behaviour are first-order error terms**: Kumpika's duty-cycling result and Le's CL61 laser-power/calibration-factor finding both describe degradation that a deployed node cannot self-detect. A co-location rotation schedule is not optional infrastructure.

Confidence that this week was informative: high — but note that the single largest change in the corpus was a fix to *this pipeline*, not to the literature. Nine of the twenty sensing records exist in this rollup because a harvest leg was built on Friday. The literature did not change; the instrument reading it did.

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Cai et al.	$PM_{2.5}$ as the dominant air pollutant associated with suicide among people living with HIV/AIDS: province-wide case-crossover	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	China
Calderon-Garcidueñas	Abnormal eye movements reflect early cortical and volumetric brain changes in $PM_{2.5}$-exposed urban youth	*J Alzheimers Dis*	Cross-sectional imaging	$PM_{2.5}$	Mexico
Cattarinussi	Independent and joint effects of outdoor air pollution and genetic risk on adolescent mental health trajectories	*medRxiv*	Prospective cohort	$PM_{2.5}$	USA
Dales	Wildfire-sourced $PM_{2.5}$ and ED visits for migraine and other primary headache	*JAMA Netw Open*	Case-crossover	$PM_{2.5}$ (wildfire)	Canada
Daoud et al.	Air pollution and suicide mortality in the UK: implications for environmental and mental health policy	*Environ Sci Pollut Res Int*	Ecological panel, fixed effects	$PM_{2.5}$ (modelled, DEFRA)	UK
Glover	Umbrella meta-review of urban environmental exposures and cognitive health	*Age Ageing*	Umbrella review / meta-analysis	$PM_{2.5}$	Global
Goossens	In utero air pollution exposure and pre-/postnatal brain development	*Neurosci Appl*	Narrative review	$PM_{2.5}$ / PM_{10} / PM_{coarse}	Global
Kabir	Neighbourhood socioeconomic inequalities in mental health: moderation by environmental attributes	*Health Place*	Scoping review	PM (unspeciated) / emissions	Global
Lu	Subchronic $PM_{2.5}$ exposure exacerbates chronic stress-induced depression-like pathology in mice	*Environ Res*	Murine inhalation exposure	$PM_{2.5}$	China
Ma	$PM_{2.5}$-induced neurotoxicity and depression: exposure risks and mechanisms	*iScience*	Narrative review	$PM_{2.5}$ + components	Global
Pan	Maternal PM exposure and offspring neurodevelopment: effects and mechanisms	*Zhonghua Yu Fang Yi Xue Za Zhi*	Narrative review	PM (bulk)	China
Sarkar	Particulate-matter-induced neurotoxicity: neuroinflammation and organelle dysfunction	*J Appl Toxicol*	Narrative review	$PM_{2.5}$ + PM_{10} jointly	Global
Zhang et al.	Association of $PM_{2.5}$ and its components with cognitive dysfunction among rural older adults	*Neurotoxicology*	Cross-sectional (cohort baseline)	$PM_{2.5}$, BC, OM, NH4, SO4, NO3 (TAP)	China
Cardiovascular & metabolic					
Belay	Long-term bushfire-related $PM_{2.5}$ and coronary plaque burden in asymptomatic adults	*Eur J Prev Cardiol*	Prospective cohort	$PM_{2.5}$ (bushfire)	Australia
Chen	Long-term particulate matter exposure and coronary inflammation by pericoronary fat attenuation index	*Environ Pollut*	Retrospective cohort	$PM_{2.5}$ + PM_{10} jointly	China
Chrysohoou	Preventing cardiovascular disease within the built and living environment: an expert opinion	*Eur J Prev Cardiol*	Narrative review	PM (unspeciated)	EU-27
Huang	$PM_{2.5}$ and dyslipidaemia in rural Chinese children; dietary effect modification	*Zhonghua Liu Xing Bing Xue Za Zhi*	Cross-sectional (multistage)	$PM_{2.5}$	China
Kazmierska	Air pollution and cardiovascular disease: review of current data	*Przegl Epidemiol*	Narrative review	$PM_{2.5}$ / PM_{10}	Global
Luo	Short-term ambient air pollution and cardiovascular mortality: an 11-year time-series in Nanning	*Front Public Health*	Time-series	$PM_{2.5}$ + PM_{10}	China
Rusconi	Long-term ultrafine particle exposure 2010-2019 in high-income countries and cardiovascular burden	*GeoHealth*	Ecological panel	Ultrafine particles	Global
Xi	Air pollution, built environment and temperature vs. prediabetes phenotypes	*Diabetologia*	Prospective cohort	$PM_{2.5}$	Germany
Yang	Short-term $PM_{2.5}$ and hospitalisation/ED use in atrial fibrillation	*J Am Heart Assoc*	Case-crossover	$PM_{2.5}$ (wildfire-influenced)	USA
Zhao	Compound exposure to extreme heat and substantial landscape fire-sourced air pollution and cardiovascular mortality	*J Am Coll Cardiol*	Case-crossover	$PM_{2.5}$ (fire-sourced)	Global
Zhou	Air pollution and hypertension: umbrella review of systematic reviews and meta-analyses	*iScience*	Umbrella review / meta-analysis	$PM_{2.5}$ / PM_{10}	Global
Reproductive & developmental					
Bello et al.	Prenatal exposure to $PM_{2.5}$ and infant health: evidence from Quebec	*Econ Hum Biol*	Registry cohort, FE models	$PM_{2.5}$ (monitor-assigned)	Canada

First author	Focus	Journal	Design	Metric	Region
Chen	Prenatal air pollution, fetal growth trajectory phenotypes and child neurodevelopment	*J Environ Sci (China)*	Prospective cohort	$PM_{2.5}$ / PM_{10}	China
Huang	Joint prenatal bisphenol + air-pollutant mixture exposure and preschool IQ	*Ecotoxicol Environ Saf*	Prospective cohort (mixtures)	$PM_{2.5}$ / PM_{10}	China
Li	Ambient air pollutant exposure and semen quality: a refined multi-window panel study	*Ecotoxicol Environ Saf*	Panel study	$PM_{2.5}$ + PM_{10}	China
Lin	Maternal particulate matter exposure during pregnancy and offspring liver function	*Environ Pollut*	Prospective cohort	$PM_{2.5}$ + PM_{10} jointly	China
Liu	Deoxynivalenol, cytokines and particulate matter in human trophoblast cell models	*Int J Mol Sci*	In vitro (cell lines)	$PM_{2.5}$	Netherlands
Pai et al.	Household incense burning and small for gestational age risk in newborn infants	*Pediatr Neonatol*	Prospective birth cohort	Incense PM (indoor, unspeciated)	Taiwan
Peterson	Ambient air pollution, PAHs and pubertal development: critical review	*Curr Environ Health Rep*	Critical review	$PM_{2.5}$ components	Global
Sezer et al.	Preconception and early pregnancy exposure to ambient air pollution and risk of cleft lip and/or palate	*Birth Defects Res*	Case-control	$PM_{2.5}$ + PM_{10} + O3	Turkiye
Xie	Long-term $PM_{2.5}$ and timing of pubertal development in boys and girls	*Environ Res*	Prospective cohort	$PM_{2.5}$	China
Respiratory & allergic					
Bayar Muluk	Climate change and otolaryngology: airway disease impacts and emission reduction	*Int J Biometeorol*	Narrative review	$PM_{2.5}$	Global
Chaiwong	$PM_{2.5}$ and systemic inflammatory biomarkers in elderly COPD patients versus healthy elderly	*Int J Mol Sci*	Panel study	$PM_{2.5}$	Thailand
Ghana AIRCLARE	Short-term ambient $PM_{2.5}$/PM_{10} and acute respiratory morbidity in Ghana: time-stratified case-crossover pilot	*Research Square (preprint)*	Case-crossover	$PM_{2.5}$ / PM_{10}	Ghana
Li	Human genetic variation, airway microbiome, environmental exposure and respiratory health	*mSystems*	Prospective cohort	$PM_{2.5}$ only	China
Liao	Short-term air pollution, respiratory symptoms and lung function in asthma	*Med Gas Res*	Panel study	$PM_{2.5}$ / PM_{10}	China
Lu et al.	Antagonistic effects of $PM_{2.5}$ and ozone on respiratory morbidity: multi-pollutant interaction, Pearl River Delta	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	China
Mokoena	Socio-economic conditions, air pollution sources and eczema symptoms among teenagers in Soshanguve	*Public Health Chall*	Cross-sectional (multistage)	PM (unspeciated) / emissions	South Africa
Sanchez de Prada	Ambient PM and asthma-coded admissions in children 0-3 y, Spain	*Epidemiologia*	Ecological panel	$PM_{2.5}$ / PM_{10}	Spain
Wang	Environmental and socioeconomic influences on childhood asthma and pneumonia burden	*Front Med*	Retrospective cohort	$PM_{2.5}$	China
Yang	Allergenic vs. toxic risk of airborne seed hairs compared with ambient $PM_{2.5}$	*J Hazard Mater*	Field measurement	$PM_{2.5}$ + bioaerosol	China
Yue et al.	Short-term traffic-related gaseous pollutant exposure and childhood bronchopneumonia hospitalization in Guangzhou	*Front Public Health*	Time-series	Traffic gases (NO2, SO2)	China
Zhang	Air pollution exposure in infancy and incidence of atopic dermatitis in preschool-aged children	*Epidemiology*	Prospective cohort	$PM_{2.5}$ only	USA
Zhu et al.	Air pollution and pneumonia hospitalization in Qingyang, China: a time-series analysis	*Medicine (Baltimore)*	Time-series	$PM_{2.5}$ + PM_{10} + 4 gases	China
Mechanistic toxicology					
Balu et al.	SRM 1650b administration to isolated rat heart aggravates ischemia-reperfusion injury via mitochondrial dysfunction	*J Biochem Mol Toxicol*	Ex vivo perfused organ	Diesel PM (SRM 1650b)	India
Ding	$PM_{2.5}$ promotes vascular calcification via miR-27b-5p/HES1 signalling	*J Environ Sci (China)*	In vitro + organoid + murine	$PM_{2.5}$	China
Fakfum	Seasonal $PM_{2.5}$ exposure and plasma metabolome changes related to metabolic syndrome	*Toxics*	Repeated-measures biomarker	$PM_{2.5}$	Thailand
Liu	Resveratrol ameliorates $PM_{2.5}$-aggravated IBS-D via gut bacteria and hypoxanthine/linoleic acid metabolism	*Environ Toxicol Pharmacol*	Murine inhalation exposure	$PM_{2.5}$ only	China
Lu	Toxicity of $PM_{2.5}$ from incomplete solid-fuel burning in Caenorhabditis elegans	*Toxics*	Controlled exposure (ecotox)	$PM_{2.5}$ (biomass)	China

First author	Focus	Journal	Design	Metric	Region
Maraschi	Osmoregulatory disruption and metal bioaccumulation from settleable atmospheric PM	*J Comp Physiol B*	Controlled exposure (ecotox)	Settleable APM	Brazil
Mishra et al.	Aerosol oxidative potential and reactive species predicted with a chemical kinetics model	*Atmos Chem Phys*	Kinetic model + field validation	PM composition, OP-DTT / OP-AA	France / UK
Pan et al.	Effects of air pollution exposure on nasal mucosal immune-inflammatory markers in animal models of allergic rhinitis	*Front Pharmacol*	Umbrella review / meta-analysis	$PM_{2.5}$ / PM_{10} / DEP (animal)	Global
Saygin et al.	Source identification and seasonal change of trace metals, protein content and oxidative potential of fine and coarse particulates after the Kahramanmaras earthquakes	*Atmos Pollut Res*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	Turkiye
Wang	PM exposure, mitochondrial stress and extracellular matrix remodelling in scleral fibroblasts	*Exp Eye Res*	In vitro + organoid + murine	PM (unspeciated) / emissions	China
Wu	Macrophage NLRP3 inflammasome in real-ambient PM-induced hepatic glucose dysregulation	*J Environ Sci (China)*	Real-ambient murine + co-culture	Real-ambient PM	China
Occupational & indoor					
Chen	Nested suction instrument to cut intraoperative surgical-smoke $PM_{2.5}$	*Front Med Technol*	Bench / device experiment	$PM_{2.5}$ (surgical smoke)	China
Cole	Determinants of respirable crystalline silica exposure in Australian tunnelling	*Ann Work Expo Health*	Observational exposure study	Respirable dust / RCS	Australia
Ding	Stage-resolved hair Zn-isotope dynamics as a firefighting combustion-exposure tracer	*J Hazard Mater*	Repeated-measures biomarker	Combustion-derived PM	China
Ghosh	Respiratory quality of life and pulmonary function among traffic police in eastern India	*Indian J Occup Environ Med*	Cross-sectional (multistage)	PM (unspeciated) / emissions	India
Gwak	Size-resolved modelling of airborne viral RNA copies in indoor environments	*Risk Anal*	Physical exposure model	Bioaerosol / coarse	Global
Komonnithiphong	$PM_{2.5}$-bound PAHs and carcinogenic risk in charcoal barbecue (Moo Kratha) restaurants	*Toxics*	Field measurement	$PM_{2.5}$ components	Thailand
Park et al.	Environmental evaluation of wood-based remodeling in aging buildings: indoor air quality and life-cycle carbon	*Sci Total Environ*	Field measurement	PM (indoor, unspeciated)	South Korea
Qi	Emission rates of respirable crystalline silica and dust from engineered stone fabrication	*Ann Work Expo Health*	Bench / device experiment	Settleable / respirable dust	Global
Ramsden	Xenobiotic hazards in aircraft cabin air	*J Xenobiot*	Review + risk economics	Ultrafine particles	Global
Subway PSD	Characterisation and source apportionment of PM and CO2 in subway stations with differing platform door forms	*J Environ Sci (China)*	Field measurement	$PM_{2.5}$ / PM_{10}	China
Yu et al.	Indoor particulate matter and associated microplastic exposure in the educational environment of Shenzhen	*Atmos Pollut Res*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	China
Zhou L et al.	Indoor burning of Arabian incense generates abundant ultrafine particles with strong oxidative potential	*Atmos Chem Phys*	Chamber emission characterisation	UFP number + PM mass, DTT/cellular OP	Chamber (Gulf practice)
van Rheenen et al.	Microbial contamination of air conditioning and humidifying systems in hospitals and the built environment	*J Hosp Infect*	Mapping review	Bioaerosol (bacteria, fungi, protozoa)	Multi-country
Sensing, forecasting & instrumentation					
Aher	Comment on advancing air pollution forecasting: operational challenges beyond predictive accuracy	*Environ Sci Pollut Res*	Commentary / methods letter	PM (unspeciated)	India
Ainembabazi	Trends in atmospheric pollution research in Uganda, 1990-2025: a scoping review	*Toxics*	Scoping review	$PM_{2.5}$ / PM_{10}	Uganda
Feng	Calibration strategies and measurement frameworks for low-cost particulate matter sensors: a review	*Meas Sci Technol*	Critical review	$PM_{2.5}$ only	Global
Ginsburg	Global and regional calibration of low-cost $PM_{2.5}$ sensors (PurpleAir): transferability of US-based corrections	*Aerosol Sci Technol*	Field measurement	$PM_{2.5}$ only	Global
Huda	Low-cost sensor network for wildfire monitoring and air quality at remote community locations	*J Air Waste Manag Assoc*	Field measurement	$PM_{2.5}$ only	Canada
Kumpika et al.	Correction of humidity and sensor aging effects in low-cost $PM_{2.5}$ light-scattering sensors	*Environ Pollut Bioavailab*	Field co-location + chamber	$PM_{2.5}$ (PMS7003 vs BAM)	Thailand
Le et al.	Operational performance of the Vaisala CL61 ceilometer for atmospheric profiling	*Atmos Meas Tech*	Multi-instrument field evaluation	Attenuated backscatter, depolarisation ratio	Finland / Germany

First author	Focus	Journal	Design	Metric	Region
Mohapatra et al.	IoT-machine learning proof-of-concept framework for real-time urban particulate matter prediction	*IEEE Access*	Proof-of-concept deployment	PM1 / $PM_{2.5}$ / PM_{10} (PMS7003)	India
Msigwa	AIoT real-time sensing and calibration of low-cost PM sensors using the TCM network	*IEEE Internet Things J*	Deep-learning model	$PM_{2.5}$ only	Global
Nurseitov et al.	Design, deployment and field evaluation of a low-cost IoT monitoring network for urban particulate matter	*Information*	Network deployment + field evaluation	$PM_{2.5}$ / PM_{10} (ZPHS01B)	Kazakhstan
Park et al.	Diagnosing explainability of machine learning: transformer-based $PM_{2.5}$ estimation in Seoul	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	Republic of Korea
Patel	Scalable air-quality sensor placement via gradient-based mutual information maximization	*Proc AAAI*	Deep-learning model	$PM_{2.5}$ only	Global
Patton	Human health risk of particle exposure in outdoor environments in southwestern Pennsylvania	*Environ Sci Pollut Res*	Real-time sensor campaign	$PM_{2.5}$ only	USA
Qian et al.	Enhancing accuracy of indoor air quality sensors via automated machine learning calibration	*Atmos Meas Tech*	Chamber co-location, AutoML calibration	$PM_{2.5}$ (two LCS models vs reference)	Chamber
Reis	Low-power embedded sensor node for environmental monitoring with on-board machine-learning inference	*Sensors*	Simulation (system-level)	PM + gas, multi-parameter node	Simulation
Shi	Adaptive spatiotemporal graph transformer for multi-site multi-step $PM_{2.5}$ forecasting	*J Environ Manage*	Deep-learning model	$PM_{2.5}$	China / India
Silva et al.	Low-cost sensors for indoor air quality monitoring: systematic review of accuracy, applications and limitations	*J Air Waste Manag Assoc*	Systematic review (PRISMA)	PM (OPC) and CO2 (NDIR)	Multi-country
Tsuchiya	Seasonal and indoor-outdoor characteristics of fluorescent aerosol particles in offices	*Biosensors*	Real-time sensor campaign	Bioaerosol / coarse	Japan
Wang	Joint monitoring and early warning of SARS-CoV-2 in outdoor $PM_{2.5}$ and wastewater	*Viruses*	Environmental surveillance campaign	$PM_{2.5}$ (bioaerosol)	China
Zheng et al.	Augmented transformer framework for $PM_{2.5}$ chemical composition via one-step LDI single-particle aerosol mass spectrometry	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	China
Exposure assessment & modelling					
Blanco et al.	Influence of on-road mobile monitoring design on ultrafine particle exposure models and cognitive health inferences	*J Expo Sci Environ Epidemiol*	Design resampling + cohort analysis	UFP particle number concentration	United States
Chen	Dual-isotope evidence for shipping contributions to atmospheric nitrate in coastal Chinese cities	*J Hazard Mater*	Source apportionment	$PM_{2.5}$	China
Chen et al.	Geospatial-based neighborhood hazards and asthma in adults: interaction with social vulnerability	*J Hazard Mater*	Exposome-wide association	$PM_{2.5}$ constituents + VOCs	USA
Chengdu NO3	Gas-phase reactions accelerate nitrate formation in spring and early summer in a Sichuan Basin megacity	*J Environ Sci (China)*	Supersite observation	$PM_{2.5}$ components	China
Delgado Peralta et al.	Improving $PM_{2.5}$ simulations in metropolitan Sao Paulo via PMF-based emission inventory adjustment	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	Brazil
Ding	Urban-rural balanced estimation framework for $PM_{2.5}$ exposure assessment across China	*Geo-spat Inf Sci*	Deep-learning model	$PM_{2.5}$ only	China
Hu	Fusion of ground monitoring, satellite observation and chemical transport modelling for seamless hourly 1-km $PM_{2.5}$	*Geo-spat Inf Sci*	Deep-learning model	$PM_{2.5}$ only	China
Nguyen	Satellite-driven prediction of $PM_{2.5}$: machine learning and explainable AI	*Eng Res Express*	Critical review	$PM_{2.5}$ only	Global
Nyoni et al.	Performance of road-traffic-based exposure proxies against personal $PM_{2.5}$ measurements in three Sub-Saharan African countries	*Air Qual Atmos Health*	Proxy validation vs personal exposure	$PM_{2.5}$ (personal, filtered)	The Gambia / Kenya / Mozambique
Pajak	Toxicological legacy of PAHs from an urban tire fire: soil contamination and cancer risk	*Toxics*	Dispersion model + soil sampling	PM_{10} / PAH	Poland
Patwary	Spatio-temporal assessment of PM_{10} and meteorological correlates in Cumilla, Bangladesh	*Res Sq (preprint)*	Field measurement	PM_{10}	Bangladesh
Raj	Spatiotemporal ambient air quality across urban, tourist and industrial Himachal Pradesh	*Res Sq (preprint)*	Field measurement	$PM_{2.5}$ + PM_{10}	India
Saeedi et al.	Feature selection and cross-validation strategy in land-use regression for mobile-monitored ultrafine particles	*Environ Sci Technol*	LUR model comparison	Ultrafine particle number (mobile)	Canada

First author	Focus	Journal	Design	Metric	Region
Sanchez	Morbidity related to level-two eruption of Popocatepetl in communities near the crater	*Indian J Occup Environ Med*	Ecological panel	$PM_{2.5}$ + PM_{10} jointly	Mexico
Solorzano-Araque	Multiscale exposure modelling: LOTOS-EUROS + mobility surveys + particle toxicity	*GeoHealth*	CTM + mobility integration	$PM_{2.5}$ / PM_{10}	Colombia
Zhang	High-resolution aerosol liquid water content across the contiguous US using machine learning	*npj Clim Atmos Sci*	Machine-learning model	$PM_{2.5}$	USA
Zhou M et al.	Surface $PM_{2.5}$ air pollution in 2022 India: emission updates, WRF-Chem evaluation, and source attribution	*Atmos Chem Phys*	CTM simulation + source attribution	$PM_{2.5}$ (WRF-Chem, speciated)	India
Burden, policy & mitigation					
Abi Nader	Short-term respiratory and skin disease risks from air pollution during the 2024 Beirut conflict	*Disaster Med Public Health Prep*	Environmental surveillance campaign	$PM_{2.5}$	Lebanon
Faggi et al.	Heatwaves and particulate matter increases: an Italian case study	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	Italy
Hu	Older-adult life expectancy in China: Bayesian mortality estimation and spatially heterogeneous determinants	*Front Public Health*	Bayesian + geospatial regression	PM_{10}	China
Karamzad	Ischaemic heart disease burden in MENA 1990-2021; PM attributable fraction 28.2%	*Health Sci Rep*	GBD secondary analysis	Ambient PM	MENA
Kraus	Development and evaluation of a risk-based air quality index for Germany	*Int J Hyg Environ Health*	Bayesian + geospatial regression	$PM_{2.5}$ + PM_{10} jointly	Germany
Li	Burden trends of Alzheimer disease and other dementias in China, 1990-2023	*Ann Acad Med Singap*	GBD secondary analysis	PM (unspeciated)	China
Makkar	Fungal biotechnology for air-pollution mitigation: mechanisms and engineered systems	*Environ Sci Pollut Res*	Critical review	$PM_{2.5}$	India
Qian et al.	Stronger actions are needed to advance $PM_{2.5}$ mitigation and its health benefits under population aging in China	*J Hazard Mater*	Machine-learning model	$PM_{2.5}$	China
Rees	HEPA filtration to reduce respiratory infection in care homes	*PLoS One*	RCT process evaluation	Filtered PM	UK
Shan	Policy-driven emission reductions and post-pandemic rebound, high-resolution inventories	*J Environ Sci (China)*	Emission inventory	PM emissions	China
Sun	Clarifying causal and translational inference in studies of ambient $PM_{2.5}$ and lung cancer	*Br J Cancer*	Commentary / methods letter	$PM_{2.5}$	Global
Vellaiyan	Diesel NOx emissions and public health: source-pathway-receptor synthesis	*J Environ Health Sci Eng*	Narrative review	PM (unspeciated)	Global
Zhou	Sources of ultrafine particles during severe haze periods: a spatial Durbin analysis	*Toxics*	Spatial econometric model	Ultrafine particles	China
Other clinical endpoints					
Dogan Karatas	Air pollution and sudden sensorineural hearing loss: nine-year retrospective study	*J Clin Med*	Retrospective clinical series	PM_{10} / $PM_{2.5}$	Turkiye
He	$PM_{2.5}$-bound chlorinated paraffins and sleep disorders in youth, with body weight as mediator	*Toxics*	Cross-sectional (multistage)	$PM_{2.5}$ components	China
Ion-Andrei	Chronic air pollution exposure and the immunomolecular profile of lung cancer	*Biomedicines*	Retrospective cohort	PM_{10} (with NO2/NOx)	Romania
Knapova et al.	Daily relationship between air pollution and physical activity in an industrial region	*Appl Psychol Health Well Being*	Panel study	PM_{10}	Czech Republic
Lupita	$PM_{2.5}$, preventive-care capacity and caries/edentulism burden across the EU	*Dent J*	Ecological panel	$PM_{2.5}$	EU-27
Ouyang	Independent and joint effects of short-term air pollutant exposure on eczema in two Chinese cities	*J Hazard Mater*	Time-series	$PM_{2.5}$ + PM_{10}	China
Park	Long-term PM exposure and risk of androgenetic alopecia	*JAAD Int*	Nationwide cohort	$PM_{2.5}$ / PM_{10}	South Korea
Sun	Environmental exposure and area deprivation vs. paediatric hydrocephalus outcomes	*J Neurosurg Pediatr*	Registry cohort + geospatial	$PM_{2.5}$ (satellite-derived)	USA
Zhou et al.	Environmental exposome study for prostate cancer in elderly men from Western China	*BMJ Open*	Case-control	$PM_{2.5}$ + SO2	China
Zurita	Childhood leukaemia and proximity to benzene emission sources in the Mexico City area	*Pediatr Blood Cancer*	Ecological panel	PM (unspeciated)	Mexico

Provenance and method

This rollup is an **aggregation only**. It pools `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over 27 July 2026–1 August 2026 via `PMRW_START`/`PMRW_END` and **re-queries no API**, so every record was screened and deduplicated in the daily issue that first carried it. Per-record provenance, source-query windows, exclusions and known metadata defects are documented in those daily issues and are not restated here.

Daily issues pooled: 27, 28, 29, 30, 31 July and 1 August 2026 — six consecutive issues, no gap.

Why the window starts on 27 July and not 26 July. The weekly cadence is Sunday-to-Saturday, so the nominal window is 26 July–1 August. The 26 July slot in `state/corpus/` does not hold a 26 July issue: it holds the **1–26 July retrospective backfill** (164 records, entry window `2026-07-01 -> 2026-07-26`) harvested for the July monthly rollup. Pooling it here would put a month of records inside a week, inflate every count by a factor of 2.3 and double-count material already published in `Reports/monthly/PM-Research-Watch-Monthly_2026-07-31.pdf`. It is therefore excluded and the pooled window is stated as 27 July–1 August throughout. The 164 backfill records are not lost — they are in the July monthly in full.

Effect estimates are transcribed from abstracts and use heterogeneous exposure contrasts — see the forest-plot caption; **no meta-analytic summary is computed and no I^2 is reported**, because the estimates are not exchangeable. Trend and inflection statements are judgements over a six-day window and are not statistical tests. The "where this is heading" box is explicitly judgement and is labelled as such.